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MILITARY SPECIFICATION

TEST SET, EXERCISE HEAD, MARK 485 MOD 0
(For Torpedo Mk 46 Mod 0 through 5)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Test Set, Exercise Head, Mark 485 Mod 0, hereinafter referred to as the "test set."

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified (see 6.2), the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-636	Boxes, Shipping, Fiberboard

MILITARY

MIL-S-901	Shock tests, HI (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for
MIL-E-16400	Electronic, Interior Communication and Navigation Equipment, Naval Ship and Shore: General Specification for

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167-1	Mechanical Vibration of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, Maryland 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (Attn: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified (see 6.2), the issues are those cited in the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMMAND (CAGE Code Ident 10001)

DL2541000	Exercise Head Test Set
LD546960	Test Set, Exploder Mechanism Torpedo, Mk 440 Mod 0

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified (see 6.2), the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issues of the documents cited in the solicitation.

PUBLICATIONS

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Traiff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Basic requirements.

3.2.1 Materials and treatments of materials and parts. Unless otherwise specified in the contract or purchase order (see 6.2), all materials and treatments of materials and parts used in the manufacture of the exercise head test set shall be in accordance with the requirements specified herein, and with the applicable documents listed in DL2541000.

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3.3 Performance requirements and physical characteristics.

3.3.1 Input power. The test set shall meet the performance requirements specified herein when 115.0 ± 11.5 volts alternating current (vac), at 60 ±3 Hertz (Hz), is supplied to connector pins J1-A and J1-B. The test set power source shall provide 300 watts continuously.

3.3.2 Power circuits.

3.3.2.1 Input. Power at 115 vac (see 3.3.1) shall be provided to energize the 40 volt direct current (vdc) and the −40 vdc power supplies. The test set power indicator lamp (red) shall illuminate when test set power switch is placed to on. The 40 vdc or −40 vdc power supplies shall provide all dc power required for test set functions.

3.3.2.2 Outputs. The dc voltages shall be connected to outputs of the test set. Exercise head power indicator lamp shall illuminate only when the exercise head power switch is on.

3.3.2.3 Oscillograph lamp power. An oscillograph lamp switch shall provide 40.0 ±2.0 vdc to oscillograph lamp output in on position.

3.3.2.4 Oscillograph motor power. An oscillograph motor switch shall provide −40.0 ±2.0 vdc to oscillograph motor output in the on position.

3.3.3 Internally supplied voltages

3.3.3.1 Direct current voltages. The test set shall supply dc voltages of 40 vdc and −40 vdc. The dc voltages shall be regulated within ±0.2 vdc from no load to maximum load (40 ohms).

3.3.3.2 Alternating current voltages. The test set shall supply ac voltages of 5 volts root mean square (vrms), 26 vrms, and 32 vrms, all at a frequency of 1850 ±20 Hz. The 5 vrms and 26 vrms shall be regulated within ±5 percent from no load to maximum load (4700 ohms). The 32 vrms shall be regulated within ±4 vrms from no load to full load.

3.3.4 Test functions.

3.3.4.1 Voltage monitoring. The voltage monitor meter shall monitor internal voltages when the exercise head power switch is in the off position and external voltages when the exercise head power switch is in the on position. Voltage monitor indications shall be in accordance with table I.

TABLE I. Voltage monitor indications.

Voltage select switch position	Voltage source	Meter indication
40	Internal	40 on 50 volt scale
	External	40 on 50 volt scale
−40	Internal	40 on 50 volt scale
	External	40 on 50 volt scale
−10	External	10 on 15 volt scale
Bat. 1	External	2.65 on 5.0 volt scale
Bat. 2	External	2.65 on 5.0 volt scale

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3.3.4.2 Oscillograph test circuitry.

3.3.4.2.1 Oscillograph signal selection. Sequence of switching and signal outputs shall be as specified in table II with course gyroscope switch positioned as indicated.

TABLE II. Oscillograph input switching sequence.

Oscillograph Galvanometer input select switch position	Simulated signal	Signal voltage
1A	Valid echo	−12.0 ±0.6 vdc
1B	Course gyro (46-0/1)	5.00 ±0.25 vrms
	Course gyro (46-2)	0.385 ±0.023 vdc
2A	Rpm ind mod "0"	−5.00 ±0.25 vdc[1]
2B	Rpm ind mod "1"	32.0 ±4.0 vrms
3A	Range gate	−12.0 ±0.6vdc
3B	Amplitude threshold	−12.0 ± 0.6 vdc
5A	Right command	5.00 ±0.25 vdc
5B	Left command	−5.00 ±0.25 vdc
5C	+K	−12.0 ±0.6 vdc
6A	Down command	5.00 ±0.25 vdc
6B	Up command	5.00 ±0.25 vdc
6C	−K	−12.0 ±0.6 vdc
6D	Valid echo inhibit	−12.0 ±0.6 vdc
7A	90 V inst (SRA)	40.0 ±2.0 vdc
7B	90 V inst (Inertia)	40.0 ±2.0 vdc
7C	Exploder arm	−40.0 ±2.0 vdc
8A	Target angle threshold	−12.0 ±0.6 vdc
8B	Noise threshold	−12.0 ±0.6 vdc

[1]No load only-isolated through 15,000 ohms resistance

3.3.4.2.2 Oscillograph signal interrupt. A momentary switch (signal interrupt) shall be provided. This switch shall open the circuit through the oscillograph galvanometer input select switch and interrupt the signal to the oscillograph.

3.3.4.2.3 Oscillograph signal test points. Jacks shall be provided on the test set panel which shall monitor the signal selected by the oscillograph galvanometer input select switch. The jacks shall be placarded "sig" for signal and "grd" for ground.

3.3.4.3 Steering control test circuitry.

3.3.4.3.1 Steering control signals. The test set shall provide steering control signals to the exercise head. With the steering switch in the normal position, a −12.0 ±0.6 vdc signal shall be provided to (range >B), and with the steering switch in the turnaway position, a ground signal shall be provided to (range >B).

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3.3.4.3.2 Steering control indications. The steering control circuitry shall provide an 1850 ±20 Hertz (Hz), 20.5 ±2.5 volt peak-to-peak (p-p), square wave signal to electronic drive out. When a signal in-phase with the electronic drive out signal is applied to electronic drive in input, the steering-normal indicator lamp shall illuminate. When a signal 180 degrees out-of-phase with the electronic drive out signal is applied to electronic drive input, the steering-turnaway indicator lamp shall illuminate.

3.3.4.4 Squib fire test circuitry.

3.3.4.4.1 Power failure squib. A power failure indicator lamp shall be provided on the test set panel. The power failure indicator lamp shall illuminate when a 40, 0, −20 vdc signal is applied to the power failure squib input of the test set.

3.3.4.4.2 Depth cutoff (Dco) squib. A depth cutoff (Dco) indicator lamp shall be provided on the test set panel. The Dco indicator lamp shall illuminate when a 2.7 ± 0.3 vdc signal is applied across two Dco squib inputs of the test set.

3.3.4.4.3 Termination squib. A termination indicator lamp shall be provided on the test set panel. The termination indicator lamp shall illuminate when a 2.7 vdc signal is applied across the term squib input and the term squib return (ret) input of the test set.

3.3.4.5 Snake end indication. A snake end indicator lamp shall be provided on the test set panel. The snake end indictor lamp shall illuminate when a ground is provided at the 3 minimum (min) snake input of the test set.

3.3.4.6 Recovery indication. A recovery indicator lamp shall be provided on the test set panel. The recovery indicator lamp shall illuminate when a ground is provided at the recovery signal input of the test set.

3.3.4.7 Inertia switch signal. A momentary switch shall be provided on the test set panel which, when depressed, shall provide a 40.0 ± 2.0 vdc signal to the inertia switch output of the test set.

3.3.4.8 Continuity and resistance. Resistance between the following pins of connector J2 shall be as follows:

J2	Resistance
w to h steering (steering switch in turnaway position)	8.0 to 8.4 k ohms
w to t	1.0 ohm max
w to v	1.0 ohm max
w to l	1.0 ohm max
w to k	1.0 ohm max
a to b	1.0 ohm max
a to c	1.0 ohm max
d to z	1.0 ohm max
f to n	1.0 ohm max
j to l	1.0 ohm max
s to t (See note 1)	1.0 ohm max
v to u (See note 1)	1.0 ohm max
a to TP 1 (galvanometer input select switch in pos 1-A)	1.0 ohm max
a to TP 1 (galvanometer input select switch in pos 1-A, S-2 depressed)	open circuit

Note 1: Arming plug connected to front panel.

3.3.4.9 Arming plug interlock. An arming plug receptacle (J4805) shall be provided on the test set front panel. An exercise head test set arming plug (P4805) shall be provided and remain connected to test set arming plug receptacle (J4805) until an arming plug from an external source is to be connected to the receptacle.

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3.3.4.10 Indicator colors. The indicator lamps used in the test set shall be colored as follows:

Test set power	red
Exercise head power	red
Snake end	yellow
Recovery	green
Power failure	green
Depth cutoff (Dco)	green
Termination	green
Normal	green
Turnaway	red

3.3.5 Service life. The service life of the test set shall be 5 years normal service with a minimum of 60 days between calibrations.

3.4 Environmental requirements. The test set shall perform the product characteristics of this specification, after being exposed, at any temperature in the applicable range to the environmental requirements under all conditions specified in this section.

3.4.1 Vibration. The test set shall withstand the type I vibration required as specified in MIL-STD-167-1, except that the variable frequency and endurance test requirements shall be as follows: The test set in operating condition shall be vibrated from 5 to 15 Hz at a half peak amplitude of 0.030 ±0.006 inch and from 16 to 25 Hz at a half peak amplitude of 0.020 ±0.004 inch. The vibration shall be applied in discrete intervals of 1 Hz and applied for 1 minute at each interval (see 6.4.1).

3.4.2 Shock. The test set shall withstand the shock required in accordance with MIL-S-901, for grade B, class I, lightweight equipment type A, in the nonoperating condition, except that the height of the hammer drops shall be from 1 foot, 1-½ feet, and 3 feet (see 6.4.2).

3.4.3 Inclination. The test set shall meet the inclination requirements in accordance with MIL-E-16400.

3.4.4 Temperature. The test set shall withstand the temperatures specified in MIL-E-16400 or range 4 equipment (see 6.4.3).

3.4.5 Humidity. The test set shall withstand the humidity requirements specified in MIL-E-16400 (see 6.4.4).

3.5 Workmanship. The workmanship and finish shall be of sufficiently high grade to assure satisfactory operation consistent with the requirements of this specification. All parts used in the test set shall be thoroughly clean and free of excess materials, chips, and loose or splattered foreign materials. The standard of workmanship exhibited in any approved first article sample, subject to any qualifications stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection rth in this specification shall become a part of the contractor's overall inspection system or quality program. bsence of any inspection requirements in the specification shall not relieve the contractor of the responsibility ıuring that all products or supplies submitted to the Government for acceptance comply with all requirements ı contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain rmance to requirements; however, this does not authorize submission of known defective material, either ıted or actual, nor does it commit the Government to accept defective material.

l.2 Classification of inspections. Inspections of the test set shall be classified as follows:

a. First article inspection
   (1) First article sample (see 4.3.1)
   (2) First article tests and inspection (see 4.5)
b. Quality conformance inspection
   (1) Quality conformance sampling (see 4.6)
   (2) Quality conformance tests and examinations (see 4.6 and 4.7)

l.3 Sampling.

l.3.1 First article sample. A first article sample, manufactured in accordance with 3.1, shall be subjected ə first article tests detailed in 4.5 at an activity designated by the contracting activity. Acceptance of the first e sample shall be based on no defects in the sample. Further production of the test set by the contractor, pproval of the first article sample, shall be at the contractor's risk.

l.4 Environmental test conditions. Unless otherwise specified (see 6.2), the test set shall be subjected to under the following conditions:

a. Temperature	Room ambient [60 to 95 degrees Fahrenheit (°F)]
b. Humidity (relative)	Room ambient, up to 95 percent
c. Altitude	Normal ground atmospheric pressure

4.5 First article test and inspection. The first article test and inspection shall consist of the examinations tests as specified in 4.5.1 and 4.5.2. Prior to, and immediately after the first article tests, the test set shall be ected to the quality conformance tests and examinations (see 4.6 and 4.7).

4.5.1 First article test equipment. The following items of test equipment are required to perform the first le tests specified herein:

a. A vibration test fixture for applying vibration as specified in 3.4.1.
b. A shock fixture for applying shocks specified in 3.4.2.
c. An inclination testing device for performing the inclination tests specified in MIL-E-16400.
d. A humidity-temperature chamber for applying the humidity and temperature tests specified in MIL-E-16400.
e. Items of test equipment specified in 4.6.10.

4.5.2 First article tests.

.2.1 Vibration. Vibrate the test set in accordance with vibration schedules required in accordance with -STD-167-1 with deviations specified in 3.4.1.

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4.5.2.2 Shock. Secure the test set to the shock test fixture rigidly and apply shocks as specified in MIL-S-901 with deviations specified in 3.4.2.

4.5.2.3 Inclination. Subject the test set to the inclination tests specified in MIL-E-16400.

4.5.2.4 Temperature. Subject the test set to temperature tests, both operating and nonoperating, as specified in MIL-E-16400 for range 4 equipment.

4.5.2.5 Humidity. Subject the test set to the humidity test specified in MIL-E-16400.

4.6 Quality conformance inspection.

4.6.1 Quality conformance test and examinations. Each test set delivered under contract shall be subjected to the tests and examinations specified herein.

4.6.2 Quality conformance criteria. Failure of the end product to pass any examination or test of this section will automatically classify the unit as nonconforming. When nonconforming units are corrected by the contractor, the unit shall be reinspected. The reinspection of a nonconforming unit may be limited to the test of examination which define the nonconformance, or when so directed by the cognizant inspector, a complete re-examination and retest of the unit may be required. Nonconforming units, which have not been corrected, will be considered for acceptance only upon formal application by the contractor to the contracting activity, via the cognizant Government representative.

4.6.3 Quality conformance sampling. The test and examinations specified in 4.6 and 4.7 shall be conduct on each test set to be delivered; therefore, there shall be no sampling plan for acceptance. Defective units sha be rejected.

4.6.4 Quality conformance periodic production tests.

4.6.5 Quality conformance periodic production test equipment. The following items of test equipment, or equivalent items, will be required to perform the tests specified herein:

a. Temperature chamber for providing the environment specified in 3.4.4.
b. Vibration test fixture for applying vibration as specified in 3.4.1.

4.6.6 Quality conformance periodic production sampling. Periodic production samples shall be selected from units having satisfactorily passed quality conformance tests and examinations of 4.6 and 4.7. These samples shall be subjected to the tests of 4.6.7 at an activity designated by the contracting activity. Periodic production sampling shall be as follows:

a. Step 1. Upon initiation of production, randomly select 1 unit from each group of 20 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, shift to step 2. Each sample so selected shall be numbered consecutively starting with sample number 1.

b. Step 2. Randomly select 1 unit from each group of 50 units. Continue with step 2 for the balance of the contract or until a defective unit is found. Each sample so selected shall be numbered consecutively starting with a sample number one greater than the highest sample number selected in step 1.

4.6.7 Quality conformance periodic production tests. Each periodic production sample, after satisfactorily passing the quality conformance tests and examination (4.7), shall be subjected to environmental tests for temperat (4.5.2.4) and for vibration (4.5.2.1). Unless otherwise specified (see 6.2), inability of the sample to satisfactorily pa the quality conformance tests and examination, subsequent to these environmental temperature and vibration tests, shall be deemed a defective unit.

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4.6.8 Defective units.. When a defective unit is found in any group, acceptance of all units on hand shall be withheld and the contractor shall:

a. Determine the cause of the defect.

b. Take the necessary actions to correct or eliminate the defect from all units on hand and, if so directed by the contracting activity, from those units previously shipped.

c. Submit to the contracting activity for approval the proposed corrective action to prevent recurrence of the defect.

d. Submit to the contracting activity for approval the proposed retest program to verify that the corrective action eliminated the defect.

e. Return to 4.6.6 a. and proceed as indicated when inspection is resumed.

4.6.9 Test reports. When specified in contract or purchase order (see 6.2), a test report shall be prepared (see 6.2.2) covering the results of test specified in 4.5, first article, test 4.6 and 4.7, Quality conformance test and inspection, and forward to the contracting activity (see 6.2.1.1).

4.6.10 Quality conformance test equipment. Items of test equipment listed in table III are required to perform quality conformance tests specified herein. All items of test equipment shall be of sufficient accuracy and precision to permit valid performance of required tests. Test instruments shall be calibrated at regular intervals. Procedures for and results of all calibrations shall be subject to approval of the cognizant Government Inspector. Items having equivalent characteristics may be substituted where necessary. Verify that all test equipment used is identified as being within calibration.

TABLE III. Quality conformance test equipment.

Item	Identification
Digital DC voltmeter (range: 0.000 to 9.999 vdc; 00.00 to 99.99 vdc; input impedance: 11 megohms; Accuracy: 0.2 percent of applied voltage or ±2 digits whichever is greater.)	Non-linear systems model 451
Oscillograph (bandpass: dc to 15 megahertz; rise time: 0.023 micro-seconds; accuracy: 3 percent from 0.05 to 20 volts/cm.)	Tektronix model 535A with 53/54 preamplifier
Electronic counter (range: 10 Hz to 120 kilohertz; accuracy: ±1 count.)	Hewlett-Packard model 522B
AC VTVM (range: 1 millivolt to 300 vac; accuracy: 1 percent at full scale; frequency range: 10 Hz to 4 megahertz).	Hewlett-Packard model 400 H
DC power supply (regulated output: 0 to 30 vdc; load regulation: less than 0.3 percent, or 30 millivolt change.)	Hewlett-Packard model 721A
Multimeter (resistance range: 0 to 20 megohms; accuracy: ±3 percent of full scale.)	Simpson model 260
Load resistors 40 ohms, 5%, 40 watts 30,000 ohms, 5%, ½ watt 4700 ohms, 5%, ½ watt	Commercial

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4.7. Performance tests.

4.7.1 Initial test setup.

4.7.1.1 Control settings. Verify arming plug is connected. Place the test set panel controls to the following initial positions.

Control	Setting
Test set power	off
Exercise head power	off
Voltage select	off
Steering	turnaway
Oscillograph galvanometer input select	off
Lamp	off
Motor	off

4.7.1.2 Continuity checks. Using multimeter, check for continuity and resistance between the following pins of connector J2. The resistance between each set of pins shall be as indicated.

Multimeter connections	Indication
w to h	8.0 to 8.4 k ohms
w to t	1.0 ohm max
w to v	1.0 ohm max
w to i	1.0 ohm max
w to K	1.0 ohm max
A to B	1.0 ohm max
A to C	1.0 ohm max
D to Z	1.0 ohm max
F to N	1.0 ohm max
J to L	1.0 ohm max
S to T	1.0 ohm max
V to U	1.0 ohm max
a to TP1 (Galvanometer input select switch in Pos 1-A)	1.0 ohm max
a to TP1 (Galvanometer input select switch in Pos 1-A, S-2 depressed	Open circuit

4.7.2 Equipment turn-on.

4.7.2.1 Connect test set cable W1 to AC power connector J1 on test set panel. Connect plug of cable W1 to 115 vac, 60 Hz power outlet.

4.7.2.2 Place test set power switch on test set panel to on. Test set power indicator on test set panel shall illuminate. Allow test set to warm up for 10 minutes.

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4.7.3 Indicator lamp check.

4.7.3.1 Depress each of the following indicator/switches on the test set panel. Each indicator shall illuminate when depressed.

Snake end	Exercise head power
Power failure	Recovery
Termination	Depth cutoff (Dco)
Turnaway	Normal

4.7.3.2 Verify that each indicator is colored as follows:

Test set power	red
Exercise head power	red
Snake end	yellow
Recovery	green
Power failure	green
Depth cutoff (Dco)	green
Termination	green
Normal	green
Turnaway	red

4.7.4 Power supply test 40 volt.

4.7.4.1 Connect digital DC voltmeter to pins J2-w ground and J2-z (+) on test set panel.

4.7.4.2 Place exercise head power switch on test set panel to on position. Digital DC voltmeter shall indicate 38.0 to 42.0 vdc.

4.7.4.3 Place exercise head power switch to off position and connect 4-ohm load resistor across pins J2-w and J2-z.

4.7.4.4 Place exercise head power switch to on position. Digital DC voltmeter shall indicate 38.0 to 42.0 vdc.

4.7.4.5 Place exercise head power switch to off position.

4.7.5 Power supply test −40 volt.

4.7.5.1 Connect digital DC voltmeter to pins J2-w ground and J2-y (−) on test set panel.

4.7.5.2 Place exercise head power switch on test set panel to on position. Digital DC voltmeter shall indicate −38.0 to −42.0 vdc.

4.7.5.3 Place exercise head power switch to off position and connect 40-ohm load resistor across pins J2-w and J2-y.

4.7.5.4 Place exercise head power switch to on position. Digital DC voltmeter shall indicate −38.0 to −42.0 vdc.

4.7.5.5 Place exercise head power and test set power switches to off position and disconnect digital DC voltmeter and load resistor.

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4.7.6 Power test 1850 Hz.

4.7.6.1 Connect 30,000 ohm, 5 percent, load resistor across pins J2-AE and J2-k on test set panel to simulate load.

4.7.6.2 Connect the AC pin of the VTVM between ground test point on test set panel and pin J2-AE.

4.7.6.3 Place test set power switch to on position. The VTVM shall indicate 24.7 to 27.3 vrms.

4.7.6.4 Connect electronic counter between ground test point on test set panel and pin J2-AE. Indication shall be 1830 to 1870 Hz.

4.7.6.5 Place test set power switch to off position and disconnect all test equipment except load resistor.

4.7.7 Voltage output test.

4.7.7.1 Place test set power switch and exercise head power switch to on positions. Test set power indicator shall illuminate.

4.7.7.2 Check voltage outputs of galvanometer input circuit test set by connecting voltmeter (AC or digital DC as indicated) between ground test point on test set panel and pins of connector J2 as follows. Place oscillograph galvanometer input select switch and course gyroscope switch in position indicated in table IV. Voltmeter shall indicate as specified in table IV.

TABLE IV. Galvanometer input test.

Test no.	Voltmeter type	J2 Pin no.	Switch position	Indication
1.	DC	a	1-A	−11.4 to −12.6 vdc
2.	AC	u	1-B (46-0/1)	4.75 to 5.25 vrms
3.	DC	u	1-B (46-2)	+0.362 to +0.408 vdc
4.	DC	b	2-A	−4.75 to 5.25 vdc
5.	AC	m	2-B	28.0 to 36.0 vrms*
6.	DC	c	3-A	−11.4 to −12.6 vdc
7.	DC	AL	3-B	−11.4 to −12.6 vdc
8.	DC	n	5-A	+4.75 to +5.25 vdc
9.	DC	p	5-B	−4.75 to 5.25 vdc
10.	DC	d	5-C	−11.4 to −12.6 vdc
11.	DC	q	6-A	−4.75 to 5.25 vdc
12.	DC	r	6-B	+4.75 to +5.25 vdc
13.	DC	f	6-C	−11.4 to −12.6 vdc
14.	DC	AM	6-D	−11.4 to −12.6 vdc
15.	DC	j	7-A	+38.0 to +42.0 vdc
16.	DC	k	7-B	+38.0 to +42.0 vdc
17.	DC	g	7-C	−38.0 to −42.0 vdc
18.	DC	AN	8-A	−11.4 to −12.6 vdc
19.	DC	AP	8-B	−11.4 to −12.6 vdc

*Measured with 4700 ohms load between pin J2-m and ground.

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4.7.7.3 With digital DC voltmeter connected as in last step, depress signal interrupt switch on test set panel. Digital DC voltmeter shall indicate zero volts. When signal interrupt switch is released, digital voltmeter indication shall return to −11.4 to −12.6 vdc.

4.7.7.4 Place oscillograph galvanometer input select switch to off position.

4.7.7.5 Connect digital DC voltmeter between ground test point on test set panel and pin J2-x.

4.7.7.6 Place lamp switch to on position. Digital DC voltmeter shall indicate 38.0 to 42.0 vdc. Place oscillograph lamp switch to off position.

4.7.7.7 Connect digital DC voltmeter between ground test point on test set panel and pin J2-w.

4.7.7.8 Place motor switch to on position. Digital voltmeter shall indicate −38.0 to −42.0 vdc. Place oscillograph motor switch to off position.

4.7.7.9 Connect digital DC voltmeter between ground test point on test set panel and pin J2-h. Digital DC voltmeter shall indicate zero.

4.7.7.10 Place steering switch to normal position. Digital DC voltmeter shall indicate −11.4 to −12.6 vdc.

4.7.7.11 Place exercise head power switch to off position and disconnect test equipment.

4.7.8 Voltage monitor test.

4.7.8.1 Place voltage select switch on test set panel to 40 position. DC voltmeter on test set panel shall indicate 38.0 to 42.0 vdc. (Read indication on 50-volt scale.)

4.7.8.2 Place voltage select switch on test set panel to minus 40 position. DC voltmeter on test set panel shall indicate −38.0 to −42.0 vdc. (Read indication on 50-volt scale.)

4.7.8.3 Place voltage select switch on test set panel to off position and place exercise head power switch on test set panel to on position.

4.7.8.4 Connect jumper wire between pins J2-z and J2-V on test set panel.

4.7.8.5 Place voltage select switch on test set panel to 240 position. DC voltmeter on test set panel shall indicate 38.0 to 42.0 vdc. (Read indication on 50-volt scale.) Place exercise head power switch to off position.

4.7.8.6 Disconnect jumper wire from pins J2-z and J2-V, and connect between pins J2-y and J2-S on test set panel. Place exercise head power switch to on position.

4.7.8.7 Place voltage select switch to minus 40 position. DC voltmeter on test set panel shall indicate −38.0 to minus 42.0 vdc. (Read indication on 50-volt scale.)

4.7.8.8 Connect external DC power supply outputs to digital DC voltmeter, set DC power supply short switch to 225 MA and adjust DC power supply for an indication of 10.0 ±0.01 on digital DC voltmeter. Connect DC power supply to pins J2-w (+ lead) and J2-AD (− lead) on test set panel.

4.7.8.9 Place voltage select switch on test set panel to minus 10 position. DC voltmeter on test set panel shall indicate −9.7 to −10.3 vdc. (Read indication on 15-volt scale.)

4.7.8.10 Connect external DC power supply outputs to digital DC voltmeter and adjust DC power supply for indication of 2.70 ±0.01 vdc. Connect DC power supply to pins J2-Y (+ lead) and J2-P (− lead) on test set panel.

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4.7.8.11 Place voltage select switch on test set panel to Battery (Bat) 1 position. DC voltmeter on test set panel shall indicate 2.54 to 2.76 vdc. (Read indication on 5-volt scale.)

4.7.8.12 Connect leads of DC power supply to pins J2-Y (– lead) and J2-E (+ lead). Depth cutoff (Dco) indicator shall illuminate during this step.

4.7.8.13 Place voltage select switch on test set panel to Battery (Bat) 2 position. DC voltmeter on test set panel shall indicate –2.54 to –2.76 vdc. (Read indication on 5-volt scale.)

4.7.8.14 Place voltage select switch on test set panel to off position, and disconnect and turn off DC power supply.

4.7.9 Indicator lamp test.

4.7.9.1 Connect jumper wire between ground test point on test set panel and pin J2-AA. Snake end indicator on test set panel shall illuminate.

4.7.9.2 Disconnect jumper wire from pin J2-AA and connect to pin J2-s. Recovery indicator on test set panel shall illuminate. Disconnect jumper wire.

4.7.9.3 Connect jumper wire between pins J2-z and J2-M. Turn exercise head power switch to on. Power failure indicator on test set panel shall illuminate.

4.7.9.4 Disconnect jumper wire from pins J2-z and J2-M. Power failure indicator on test set panel shall extinguish.

4.7.9.5 Adjust output voltage of DC power supply to 2.7 volts DC and connect to pins J2-Y (+ lead) and J2-E (+ lead). Depth cutoff (Dco) indicator on test set panel shall illuminate.

4.7.9.6 Connect DC power supply to pins J2-P (+ lead) and J2-H (– lead). Termination indicator on test set panel shall illuminate.

4.7.9.7 Disconnect and turn off DC power supply.

4.7.10 Steering test.

4.7.10.1 Connect pin J2-AC on test set panel to channel A of oscilloscope (do not use probe) and input of electronic counter. Oscilloscope shall display a square wave with an amplitude of 18 to 23 volts p-p. Electronic counter shall display a frequency of 1830 to 1870 Hz.

4.7.10.2 Set oscilloscope vertical scale to 0.2 volts/cm and oscilloscope polarity switch to normal position.

4.7.10.3 Connect vertical signal output of oscilloscope to pin J2-AB on test set panel. Steering-normal indicator on test set panel shall illuminate.

4.7.10.4 Set polarity switch on oscilloscope to inverted position. Steering normal indicator on test set panel shall extinguish and steering-turnaway indicator on test panel shall illuminate.

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4.7.11 Examination of product. The test set shall be examined for conformance to the requirements specified herein and in accordance with LD546960. The presence of any one of the following defects shall constitute a failure:

a. Test set not complete; components missing
b. Components not new, evidence of reclaimed, used, or salvaged components
c. Evidence of unauthorized materials or repairs
d. Receptacles do not accept mating plugs
e. Panel inscriptions incomplete, incorrect, or illegible
f. Case not watertight
g. Continuity not in accordance with schematic
h. Test set not correctly identified
i. Test set not preserved and packaged in accordance with section 5.

5. PACKAGING

5.1 Preservation and packaging.

5.1.1 Level A and level C.

5.1.1.1 Cleaning and drying. The test set shall be thoroughly cleaned and dried by any process that will not damage the unit.

5.1.1.2 Packaging. The test set case shall serve as the package.

5.2 Packing. Packing shall be level A or C or industrial as specified (see 6.2, 6.4.5, 6.4.6 and 6.4.7).

5.2.1 Level A. The test set, preserved and packaged in accordance with 5.1, shall be overpacked in containers in compliance with PPP-B-636, type CF, Class — Weather Resistant, variety SW, Grade V3C (see 5.2.4).

5.2.2 Level C or industrial. The test set, when preserved and packaged as specified in 5.1, shall be overpacked to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with the Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation (see 6.4.6 and 6.4.7).

5.2.3 Cushioning or dunnage. The contractor shall provide sufficient cushioning or dunnage material to adequately protect equipment from damage due to shock or vibration loads encountered during handling and transportation, and to assure safe delivery to the receiving activity.

5.3 Marking.

5.3.1 Normal marking. In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Special marking. The following marking shall appear on the top and sides of each shipping container:

"DELICATE INSTRUMENT HANDLE WITH CARE"

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The test set is used in the fleet as electronic test equipment for checkout and test of Exercise Head Mk 85 Mods 2, 3, 4, and 5.

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6.1.1 Description of test set. The test set is a portable unit of electronic test equipment contained in a watertight, shock-protected combination case with a removable cover and facilities for storage of special tools and cabling. The case also provides protection from environmental deterioration while in storage and damage from handling during shipment. The test set front panel provides controls and connectors accessible for operation and interconnection of the test set when the cover is removed. The test set consists of electronic circuitry which provides proper simulated signals and loads to the exercise head unit being tested.

6.2. Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification
b. Issue of DODISS to be cited in the solication, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.1.2, and 2.2)
c. Material and treatment (3.2.1)
d. Assigned activity for first article and periodic production test (see 4.1 and 4.3.1)
e. Environmental test conditions (4.4)
f. All additional samples required shall be provided and inspected at the expense of the contractor, when such additional samples are required because of failure of any sample to pass the prescribed inspections (see 4.6.7)
g. Is test report required (see 4.6.9)
h. Number of tests sets required
i. Required level of packaging and packing (see 5.1 and 5.2).

6.2.1 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractural application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID Number	DID Title	Suggested Tailoring
4.6.9	DI-T-2072	Reports, Test	

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.2.1.1 The data requirements of 6.2.1 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previoulsy (for example, test reports).

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves

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the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions.

6.4.1 Vibration. Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude may refer to displacement or acceleration. The displacement given in one half the peak-to-peak excursion in inches or the corresponding maximum acceleration in gravity units. The frequency is given in Hertz (Hz).

6.4.2 Shock. Shock is an abrupt change in the position of a body resulting from the sudden application of a large external force of short duration. Shock is described as a pulse force in units of gravitational acceleration of duration in units of time (milliseconds[ms]), and a shock wave approximating a one-half sine wave. The time duration shall be measured between two points where two lines, intersect the zero gravitation line. These lines are tangent to the shock wave slopes.

6.4.3 Temperature. The temperature of the environment to which the equipment may be exposed, expressed in degrees Fahrenheit (°F).

6.4.4 Humidity. The relative humidity of the environment to which the test set is exposed, expressed as a percent.

6.4.5 Level A packaging. Maximum protection, designated as level A, is the level of preservation or packing required for protection of material against the most severe worldwide shipment, handling, and storage conditions. Preservation and packing so designated will be designed to protect material against direct exposure to extremes of climate, terrain, operational and transportation environments without protection other than that provided by the pack. The conditions to be considered include, but are not limited to:

a. Multiple handling during transportation and in-transit storage from point of origin to ultimate user.
b. Shock, vibration and static loading during shipment.
c. Loading on shipdeck, transfer at sea, helicopter delivery and offshore or over-the-beach discharge, to ultimate user.
d. Environmental exposure during shipment or during in-transit operations where port and warehouse facilities are limited or nonexistent.
e. Outdoor storage in all climatic conditions for a minimum of one year.
f. Static loads imposed by stacking.

6.4.6 Level C packaging. Minimum protection, designated as level C, will be used for protection of material under known favorable conditions. In general, the following criteria determine the requirements for this level of protection.

a. Use or consumption of the item at the first destination.
b. Shock, vibration and static loading during the limited transportation cycle.
c. Favorable warehouse environment for a maximum of 18 months.
d. Effects of environmental exposure during shipment and in-transit delays.
e. Stacking and supporting superimposed loads during shipment and temporary storage.

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When level C is specified, it must reference applicable Public Laws (Code of Federal Regulations) or a specific Federal or Military Specification, Standard or Instruction.

6.4.7 Industrial packaging. Industrial packaging will be acceptable for any level of protection whenever the technical design details of the package meet all conditions of the level of protection specified. Industrial packaging must provide the same level of protection against physical and environmental damage as the military package. It will be marked to the level it meets. Bulk type practices such as are used in inter and intraplant shipments or shipments to jobbers are not acceptable, unless they are usual trade practices for selected commodities.

6.5 Subject term (key word) listing.

Test equipment
Torpedo

6.5.1 Stock Number and Part Number.

National Stock Number (NSN) 4921-00-927-5836
4925-00-927-5836

Part Number 2541000

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used to indicate changes from the previous issue due to the extensiveness of the changes. Document number changed from MIL-E-82447A(OS) to MIL-T-82447B(OS) due to prime item being a "TEST SET."

Preparing activity:
Navy - OS

Project No. 4921-N002

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